



Reproducibility and variability of submesoscale frontal eddies on a broad, low-energy shelf of freshwater influence

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Abstract

A multi-decadal simulation of ocean circulation in the northern Gulf of Mexico produces strong submesoscale instabilities in the Mississippi/Atchafalaya plume fronts. The model skill in reproducing these submesoscale frontal eddies over the Texas-Louisiana shelf is assessed using simulated and observed salinity and velocity fields as a way to investigate simulation accuracy and quantify the variability of frontal eddies. The model successfully reproduces mean salinity structure observed in multi-year densely sampled CTD profiles. Variability associated with submesoscale eddies is the largest source of error in predicted salinity. On the other hand, the model is statistically able to reproduce the magnitude and characteristics of frontal eddies; metrics for eddy kinetic energy are similar between the observations and simulation, and observed horizontal salinity gradients have similar occurrence rates in the model when sampled in a manner similar to the observations. Seasonal and inter-annual variability of frontal eddies is associated with the volume of freshwater onto the shelf and wind stress. Wind stress, the highest in winter and lowest in summer, contributes to the suppression of baroclinic instability during non-summer seasons. River streamflow, highest in spring, creates strong horizontal and vertical density gradients. These strong horizontal density gradients, along with weak seasonal upwelling-favorable winds that tend to broaden the plume, are the primary factors in exciting submesoscale instabilities during summer on the Texas-Louisiana shelf. At decadal scales, streamflow, EKE, and salinity gradients have a positive correlation suggesting that long-term variability of frontal eddies may be influenced remotely by inter-annual variability in the Mississippi River outflow.

Keywords Submesoscale · Predictability · Freshwater · Salinity gradient · Continental shelf · Northern Gulf of Mexico

1 Introduction

On the continental shelf where freshwater from rivers meets seawater, plume fronts, defined as sharp horizontal density gradients between two distinct water masses, are formed. Along these fronts, frontal eddies can be generated as a result of baroclinic instability, assuming that the width of the plume is wide enough to support eddy formation (Hetland 2017). Numerous frontal eddies have been reported from

satellite images (Fig. 1) and numerical modeling (Fig. 2a) along the plume fronts on the Texas-Louisiana (TXLA) shelf, a low-energy shelf where the high volume of freshwater from the Mississippi River plays a central role in the circulation over the shelf.

The eddies on the TXLA shelf are “submesoscale,” in that the corresponding Rossby number is close to 1, in contrast to larger mesoscale eddies that are associated with a small Rossby number (Chelton et al. 2007; Munk et al. 2000). Recent studies have demonstrated that submesoscale eddies are dynamically distinct from mesoscale eddies due to the size smaller than mesoscale eddies ($O(100)\text{m}$ to $O(10)\text{km}$), high Rossby number ($Ro=\zeta/f=O(1)$) where ζ is relative vorticity and f is inertial frequency), and high vertical transport in the surface mixed layer ($\sim 10^{-4}\text{m}\cdot\text{s}^{-1}$) (e.g., Boccaletti et al. 2007; McWilliams 2016; Thomas et al. 2013). Submesoscale eddies on the shelf are also distinct from those in open water because of shallow bathymetry of the shelf seas and terrestrial influences (e.g., Jaeger and Mahadevan 2018).

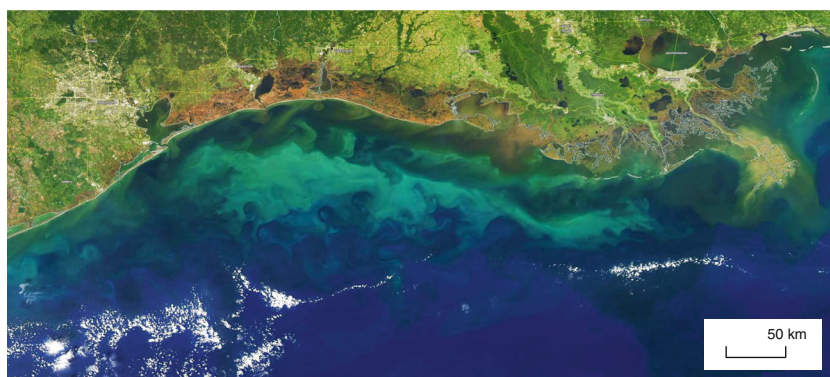
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Fig. 1 MODIS true color satellite image on September 26, 2008 (source: NASA)



On the TXLA shelf, submesoscale frontal eddies are believed to play a significant role in regional coastal circulation patterns (e.g., Zhang et al. 2012); thus, accurate prediction of eddies and understanding of their variability are vital given a potential importance of ocean circulation on economy (e.g., offshore oil exploration and potential oil spills) and the shelf ecosystems (e.g., hypoxia, harmful algal blooms). However, ocean eddies are non-linear and chaotic in nature and, thus, are known to be difficult to be reproduced exactly in numerical models, unless the models assimilate observations that resolve the eddy structure such as sea surface height or sea surface temperature from

satellites, and other in situ observations such as Argo floats. Unfortunately, data assimilation is not yet a feasible option for sub-mesoscale modeling in many parts of the ocean due to scarcity of spatio-temporal observations with the spatial resolution sufficient to resolve sub-mesoscale processes ($O(100\text{ m}–10\text{ km})$). Thus, many of the modeling efforts on submesoscale processes are limited to idealized or semi-idealized ones (e.g., Boccaletti et al. 2007; Fox-Kemper et al. 2008; Jaeger and Mahadevan 2018; Mahadevan et al. 2010).

Although data assimilation is not feasible at this moment, it is still possible to evaluate frontal eddy simulation

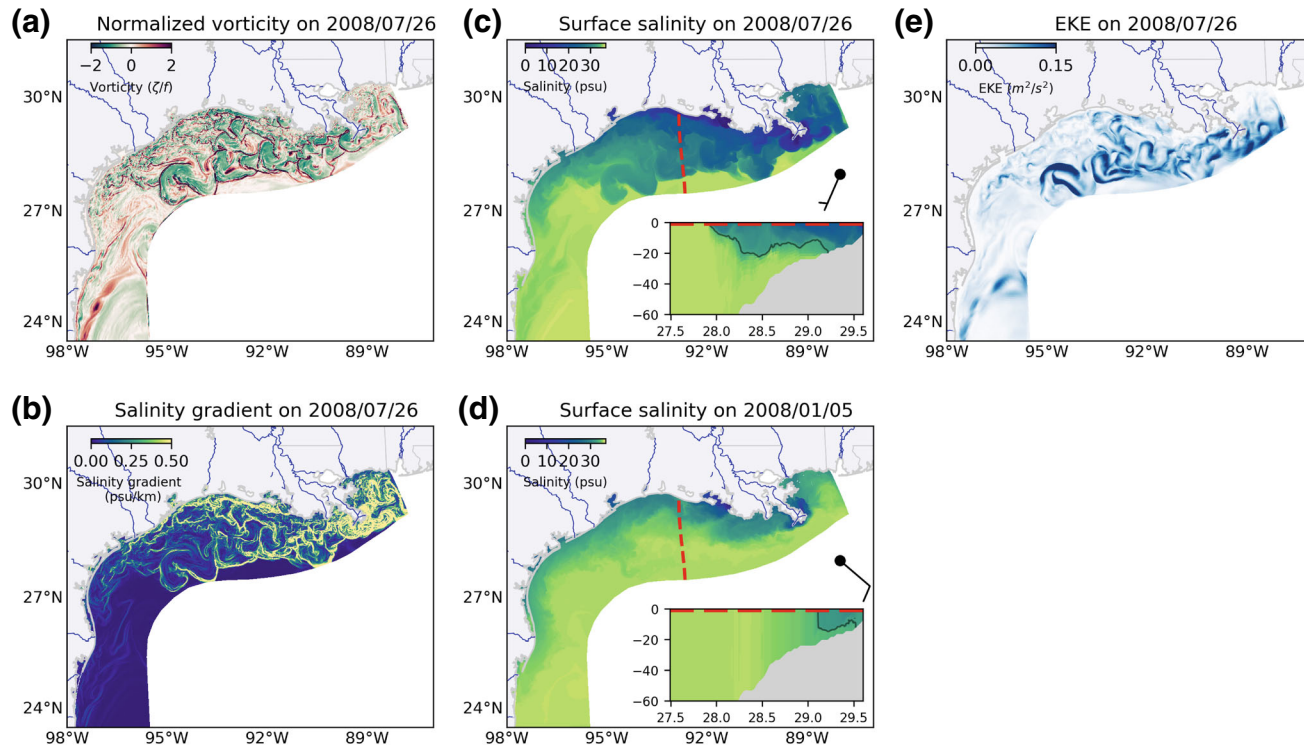


Fig. 2 Model snapshot of (a) normalized vorticity, (b) salinity gradient on 26 July 2008, (c) surface salinity on 26 July 2008, (d) surface salinity on 05 January 2008, and eddy kinetic energy on 26 July 2008.

Black line in (c) and (d) represents isohaline of $33\text{ g}\cdot\text{kg}^{-1}$. Wind bar in (c) and (d) shows wind speed and direction

based on the ability of a non-assimilative ocean model to statistically reproduce observed properties, especially if the model error largely occurs as a result of mismatches in the location of eddies, not eddy characteristics. For the TXLA shelf, periodic shelf-wide CTD measurements spanning more than a decade have been undertaken as a part of groundfish surveys by National Oceanic and Atmospheric Administration (NOAA) and have provided a multi-decadal data set. Such a data set provides an opportunity to evaluate the eddies. If the model is statistically valid for reproducing eddy structure, if not location in space and time, it should also be possible to discuss seasonal and interannual variability of the frontal eddies. There are virtually no studies on seasonal and interannual variability of submesoscale eddies on the shelf based on realistic simulations, though there are some studies for open water on seasonal variability (e.g., Mensa et al. 2013).

Here, we investigate variability of the shelf eddies over the TXLA shelf based on a multi-decadal simulation for the period between 1994 and 2016. This paper pursues three goals: (i) identify what causes model errors from eddy simulation perspectives, (ii) examine whether a non-assimilative ocean model is able to replicate frontal eddies statistically, and (iii) quantify variability of frontal eddies at seasonal and inter-annual scales and identify the processes controlling the variability. We use salinity data as well as other eddy variables to evaluate eddy simulation since the

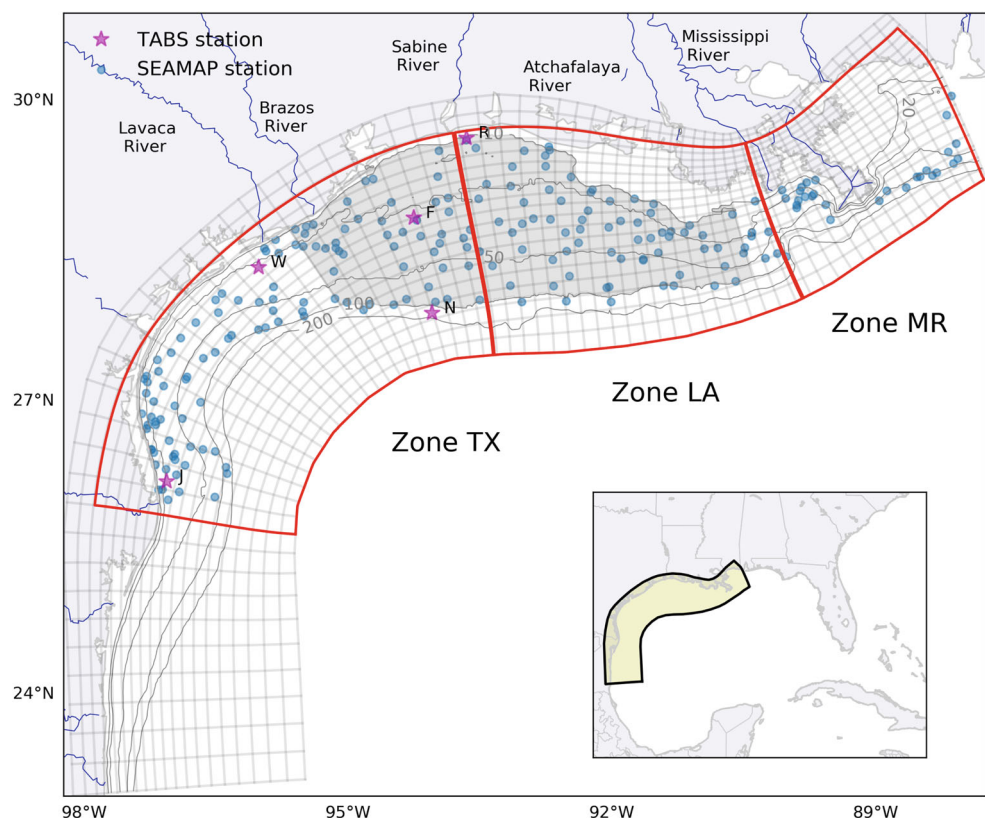
density variation on the TXLA shelf is largely dependent on salinity (Hetland and DiMarco 2012). We develop metrics to quantify the model's ability to simulate frontal eddies and discuss processes causing the free-run prediction errors associated with frontal eddies detailed in the following sections.

2 Materials and methods

2.1 Hydrodynamic model

The model configured for the TXLA shelf is based on Regional Ocean Modeling System (ROMS), which is a hydrostatic primitive equation ocean model (Shchepetkin and McWilliams 2005). The domain covers the entire Texas-Louisiana Shelf and shelf slopes (Fig. 3). The model uses a curvilinear grid stretching along the coastline of Texas and Louisiana horizontally and terrain-following stretching coordinate (s-coordinate) vertically. The finest horizontal resolution is approximately 650 m and the coarsest resolution is approximately 3.7 km. The number of vertical layers is 30 and more layers are concentrated near the surface and bottom to ensure the adequate number of layers in the surface and bottom boundary layers. As with the previous incarnations of this model configuration (e.g., Hetland and DiMatco 2012; Zhang et al. 2012),

Fig. 3 Model domain. Every 10 grids are shown. Blue dots represent SEAMAP stations for the summer 2008 cruise. The domain is divided into three zones. Gray-shaded area is the area used to estimate eddy variables and to discuss the variability. Star marks represent TABS stations



tides are not included as they are very weak over the TXLA shelf (DiMarco and Reid 1998). The shelf model is nested into global general ocean circulation models (OGCMs), Global HYCOM Reanalysis (from 1993 to 2012. <http://www.hycom.org>), and Global Mercator (from 2012 to 2016. <http://marine.copernicus.eu/>), both of which have the spatial resolution of approximately 9 km. Surface forcing and fluxes are obtained from ERA interim (Dee et al. 2011) (<https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era-interim>). A total of nine rivers, namely Sabine, San Antonio, Trinity, Brazos, Calcasieu, Lavaca, and Nueces, as well as the Mississippi and Atchafalaya Rivers, two major rivers connected to the shelf, are included. The streamflow data are obtained from U.S. Army Corps of Engineers (<https://www.mvn.usace.army.mil/>) and U.S. Geological Survey (<https://waterdata.usgs.gov/nwis>). Salinity of the rivers is set to zeros and streamflow temperature is estimated from air temperature using a bulk formulae based on Stefan and Preud'homme (1993). The model is initialized on January 1, 1993, with 1-year spin-up time and is run for the period between 1994 and 2016. For the validation of the model, the model output from 2000 and 2015 is used; whereas, for the analysis of seasonal and inter-annual variability, the model output from 1994 to 2016 is used for this study. The model uses third-order upwind scheme (horizontal) and 4th-order centered scheme (vertical) for momentum advection and MPDATA (Multidimensional Positive Definite Advection Transport Algorithm) for tracer advection (Smolarkiewicz 2006). Vertical mixing is $k-\omega$ (Warner et al. 2005) and the horizontal viscosity and diffusion coefficients are a constant value of $5.0 \text{ m}^2 \cdot \text{s}^{-1}$ and $1.0 \text{ m}^2 \cdot \text{s}^{-1}$ respectively. Both coefficients are scaled with grid size.

2.2 Field data

We evaluate simulated salinity and further frontal eddies with densely sampled CTD data collected by NOAA. Environmental profiles with a 1-m increment are acquired during the 2000–2015 Southeast Area Monitoring and Assessment Program's (SEAMAP) fall and summer groundfish surveys

of the western and north-central Gulf of Mexico. The location of each station is rather random between 5 and 100 m in depth on the shelf (Fig. 3). The shortest distance between adjacent CTD stations is approximately 500 m, which is sufficient to investigate various sizes of frontal eddies. The summer cruises are typically conducted between June and August and the fall cruises are typically conducted between October and November. The number of stations varies each year, but the number of CTD stations for each cruise over the 16-year period is roughly 150 on average (Table 1). The hydrographic stations for 2008 are illustrated in Fig. 3.

In addition, near-surface current velocity data from Texas Automated Buoy System (TABS. <http://tabs.gerg.tamu.edu>) are also used to calculate eddy kinetic energy (EKE). TABS provides a half hourly time series data for current velocities, temperature, and salinity at different locations on the shelf (see Fig. 3 on locations). The data are sub-sampled at 1-h interval in line with the model output. The sensors are located approximately 2 m below surface.

3 Results

3.1 Model skill assessment

Simulated salinity distributions are assessed using the hydrographic observations described above. There are various ways to quantify the model skill. Cross-correlation coefficient (R) and the coefficient of determination (r^2) are widely used to evaluate how close the predicted values are to the observed counterparts. Although the correlation coefficient can tell us whether simulated values are related linearly with observed values, the magnitude of the coefficient is not necessarily related to the accuracy of the prediction (Willmott 1982). Thus, it is often insufficient and misleading in using it to investigate whether the model captures certain physical patterns and characteristics such as eddies.

Since ocean circulation over the TXLA shelf is driven to a large extent by linear processes (e.g., geostrophic currents, Ekman transport, inertial currents) (e.g., Chen

Table 1 Statistics of salinity at 5 m from surface and surface EKE for each zone. Values on top are for summer and those on bottom are for fall. $\langle \cdot \rangle$ shows temporal average. $\bar{a}$ represents spatial average of variable **a**

Zone	$\sigma_{(obs)}^2$	$\sigma_{(model)}^2$	$\langle S_{err} \rangle$	$S_{err[std]}$	$\langle \frac{dS}{dt}_{err} \rangle$	$\langle \frac{dS}{dt}_{std} \rangle$	$\langle EKE \rangle$	Sample
TX	8.42	7.85	− 0.83	1.94	0.06	0.09	0.0024	1258
	5.22	5.07	− 0.62	1.64	0.05	0.14	0.0015	1043
LA	11.21	11.35	− 0.59	2.78	0.11	0.24	0.0037	1080
	4.40	5.77	− 0.51	1.43	0.04	0.08	0.0012	986
MR	14.18	11.47	0.60	3.27	0.17	0.28	0.0066	285
	5.30	4.89	0.55	1.67	0.07	0.11	0.0015	272

et al. 1996; Cochrane and Kelly 1986), the variability caused by these processes can be reproduced in the model; non-linear motions such as eddies tend to degrade the correlation coefficient, and may be incorrectly interpreted as measurement error, model error, or just noise.

Taylor Diagram (Taylor 2001) plots three statistical values, namely standard deviation, RMSD (root-mean squared deviation), and R (correlation coefficient) in a single figure and gives a more detailed picture of the model performance of salinity simulation. The salinity data used for the analysis are accumulated over the top 20 m from surface with 1-m increment. With the standard deviation normalized by the observed counterparts, we can compare multiple-year data in a single diagram since the reference value is always 1. What the Taylor Diagram can tell us is whether the model is able to reproduce similar variability as well as the same level of error and correlation.

3.2 Metrics to evaluate frontal eddies

The SEAMAP data contain randomly distributed sampling sites, and adjacent stations are close in space and time (see Fig. 3). These data provide an opportunity to calculate the changes of salinity in different distances. Note that since (i) eddies are characterized by sharp salinity gradients and (ii) the background salinity gradients ($\approx 0.02 \text{ g}\cdot\text{kg}^{-1}\cdot\text{km}^{-1}$) estimated based on the WOA (World Ocean Atlas) surface salinity gradients averaged over the TXLA shelf are about a factor of 8 lower than those at the front of the buoyant coastal current ($\approx 0.16 \text{ g}\cdot\text{kg}^{-1}\cdot\text{km}^{-1}$), we may use salinity gradients as a metric to assess model performance in predicting the frontal eddies. The following formula is used to calculate the salinity gradients from the CTD observations:

$$\left. \frac{dS}{dl} \right|_{obs} = \frac{|S_i - S_j|}{|\mathbf{x}_i - \mathbf{x}_j|} \quad (1)$$

Where $\mathbf{x}$ is the vector location of a CTD cast, S is the salinity at a certain depth, and i and j are indices representing adjacent hydrographic stations. For the purpose of comparison, we output the modeled salinity of the nearest locations and times to the sampled CTD stations and then calculate the gradients following the same formula (1) (hereafter referred to as $\left. \frac{dS}{dl} \right|_{sim.sampled}$). The corresponding results are divided into zones as defined in Fig. 3 to investigate spatial variability. Salinity gradients are presented in the form of two-dimensional histograms to evaluate the model's ability to reproduce observed small-scale salinity gradients that serve as a proxy for tracer variability due to submesoscale eddies. The histograms are normalized so that the cumulative sum is 1. We extract salinity with depths from 3 to 20 m with an increment

of 1 m and accumulate the data over 16 years (2000–2015) for analysis. The accumulation of multiple layers of data increases the sampling size and ensures robust statistical analysis. In addition, eddy kinetic energy (EKE) (Eq. 3) between the model and observations is compared statistically for several surface mooring stations, a proxy for the energy contained within submesoscale eddies.

3.3 Error analysis

There could be several sources of error in both model and field measurements, for example measurement errors, inconsistency of the locations and time stamps between the model and observations, and uncertainties in the model itself. Potential measurement errors include instrument malfunctions, errors of instruments themselves, calibration errors, and the errors by those who handle the instruments while the measurements are taken. The calibration was undertaken prior to every cruise; thus, the calibration error is likely very small. Salinity was measured by SBA-21 and SBA-45, with a nominal accuracy of about $0.01 \text{ g}\cdot\text{kg}^{-1}$. In addition, all salinity profiles were checked and outliers were removed prior to the data process. Thus, we expect errors associated with the instruments to be relatively very small.

Another potential error is bias in the numerical solution, likely caused by boundary and forcing condition errors, or errors in the numerical solution technique. However, the previous studies have shown that the model is generally unbiased (Hetland and DiMarco 2012; Marta-Almeida et al. 2013). In addition, we conducted a sensitivity test by comparing the results of the model (only 1-year run) with different numerical schemes. Though the salinity and resulting plume structure were slightly different among the different configurations, the statistical distribution of salinity gradients as described in a later part of this paper was similar among them. Our implicit assumption in this is that the errors based on the difference in sampling the simulation vs. the observations are unbiased.

Since the frequency of the model output is 1 h, the maximum time lag between the measurements and the model is $\pm 0.5 \text{ h}$ for each station. Although salinity is unlikely to change dramatically within 1 h, this is still a possible source of error. We extract simulated salinity between $\pm 1 \text{ h}$ (i.e., $-1, 0, 1 \text{ h}$) for each CTD station and calculate salinity gradients for all possible combinations (i.e., $3 \times 3 = 9$) for the depths ranging from 3 to 20 m from the surface. Then the error is compared to random noises created based on mean and standard deviation of salinity gradients. The result showed that with this method, the distribution of the error from the model was similar to the observations', and therefore the error was properly replicated as detailed in Section 3.7.

We expect that the largest potential source of error should be in mismatches between small-scale, nonlinear features that are present in the numerical solution as well as in the actual ocean state. This would be like the variance associated with turbulent flow: not predictable in detail, but the characteristic scales of variability can still be correctly predicted. Given the spread in the estimated error, we estimate this to be about $2.50 \text{ g}\cdot\text{kg}^{-1}$. Note that this error is much larger than all other estimated sources, so it should be the primary factor in model-data misfit. Thus, our task is to understand if the model is able to reproduce the structure of the nonlinear eddies correctly in a statistical sense, given it is impossible to reproduce them with strict fidelity.

3.4 Analysis of seasonal and inter-annual variability

In order to discuss seasonal and inter-annual variability of frontal eddies, surface model output is used to estimate the eddy kinetic energy (EKE), relative vorticity, and salinity gradients for the area where frontal eddies are most active (the shaded area in Fig. 3). Salinity gradients ($\frac{dS}{dl}|_{sim}$) are calculated from hourly model outputs with the following equation.

$$\left. \frac{dS}{dl} \right|_{sim} = \sqrt{\left(\frac{dS}{dx} \right)^2 + \left(\frac{dS}{dy} \right)^2} \quad (2)$$

Note that Eq. 2 is different from Eq. 1 in that salinity gradient is estimated using the model output in contrast to Eq. 1, which uses observed data to estimate the gradient. Vorticity is normalized by the inertial frequency and only positive values are used for the analysis. Frontal eddies have anti-cyclonic cores, and cyclonic vorticity along their edge. The high positive vorticity is concentrated along the perimeter of the eddies and is associated with strong fronts (Fig. 2a).

EKE is estimated by subtracting the mean current from the current velocities as follows:

$$EKE = \frac{1}{2} \{ (u - \bar{u})^2 + (v - \bar{v})^2 \} \quad (3)$$

The data are extracted for the area of intense frontal eddies in summer (shaded area in Fig. 3). The mean current ($\bar{u}$ and $\bar{v}$ in Eq. 3) is estimated by averaging the current over a period determined by the conditions outlined below. Ideally, the period to be selected meets two criteria: (i) the mean kinetic energy (MKE) does not contain EKE and (ii) the current averaged over the time period does not include seasonal currents. Previous studies have used different time-averaging windows, and it is not clear which is best in this case. In order to check which time window is best, we pick up an arbitrary date in summer (June 20) and then calculate EKE for the area between 10 and 100 m in depth with different time windows (from 2 to up to 360 days).

The process is repeated and averaged over 23 years (1994–2016). The RMS values of average EKE for different time windows indicate that at 30 days EKE reaches over 90% of maximum EKE (see Supplemental Figure S1). Therefore, we use 30-day window because the contamination of EKE in MKE is small and the influence of seasonal current is minimal, as the window is much shorter than a season.

Note that the above method is not applicable to variable wind regimes, in particular, energetic atmospheric frontal passages during non-summer, which introduce variance within the flow field that might be confounded with the EKE associated with frontal eddies. The EKE calculated from Eq. 3 contains spikes in winter time (October and May). These spikes are associated with cold fronts and are therefore considered not to be associated with submesoscale eddies we are focused on. Therefore, we consider the spikes as alias signals, and for an accurate quantification of eddy EKE, these signals need to be removed.

A simple method would be to exclude depth-averaged flow from the EKE calculation. However, this method does not entirely eliminate the EKE spikes observed in winter because the response to the wind can be depth dependent. In order to isolate the wind-driven flow, we ran a 3-D, constant density simulation using the same domain but forced only by wind, and then we subtracted the calculated EKE from the EKE calculated in the total run. We believe that by doing so, the wind-driven EKE signal can be greatly reduced, so that the new estimate for the EKE should be largely controlled by the strength of the frontal eddies.

This method is not fail-proof, in that we have no assurance that the “wind-only” EKE calculated using this method will be less than the EKE calculated from the full simulation. In practice, this was not a concern; the EKE spikes were removed and the portion of the EKE signal caused by the eddies was preserved. A number of other methods were tried, but using the “wind-only” model to subtract the wind-driven EKE signal was found to be effective.

3.5 Salinity distribution and validation

The simulated salinity distributions shown in Fig. 2c and d reveal a general pattern of salinity distribution on the TXLA shelf. Regardless of seasons, salinity increases from the east to west and also from the nearshore to offshore. There is a clear distinction in salinity distribution between summer (Fig. 2c) and non-summer seasons (Fig. 2d). Figure 2d shows an example of a typical surface salinity distribution for non-summer; there are meanders along the plume front, but no fully formed frontal eddies, and the plume is arrested in the nearshore, which is defined in this paper as the depth shallower than 20 m.

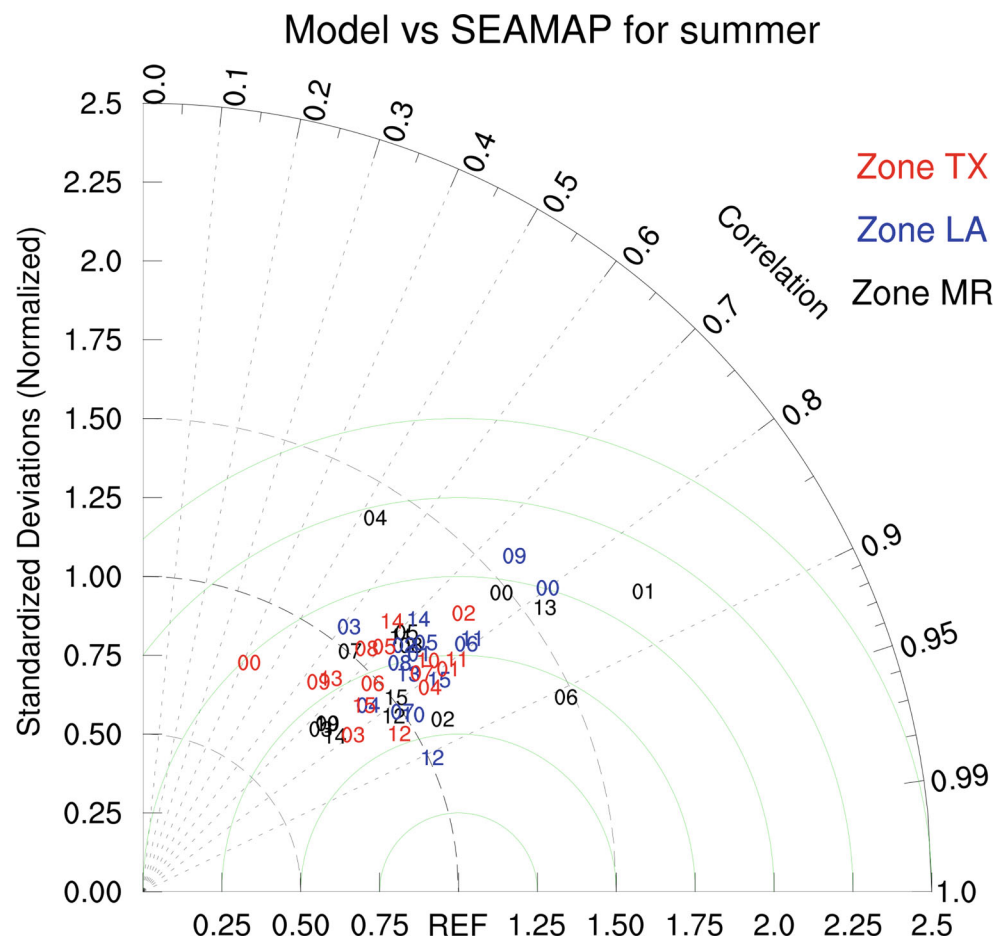
In summer, numerous frontal eddies are visible (Fig. 2c). In contrast to the narrow plume in non-summer seasons, the plume in summer is spread offshore transporting lower salinity water offshore broadening the plume width. The lower surface salinity extends to mid- and even outer shelf (the embedded figure in Fig. 2c). In summer, there is a distinct pycnocline above which salinity is low and nearly uniform in the surface mixed layer and below which salinity little varies and is close to the background deep Gulf value of $\simeq 36 \text{ g}\cdot\text{kg}^{-1}$ (see the embedded cross section in Fig. 2c). In contrast, for non-summer seasons, variability of vertical salinity is much smaller except for very near the freshwater sources, and the vertical stratification is much weaker in non-summer compared with summer seasons (see the embedded cross section in Fig. 2d). The buoyancy frequency squared (N^2) averaged over the depth of upper 20 m north of 29° North is $4.1 \times 10^{-3} \text{ (s}^{-2}\text{)}$ for summer and $1.6 \times 10^{-4} \text{ (s}^{-2}\text{)}$ for winter, respectively.

Table 1 summarizes spatial variability of salinity and various eddy properties based on the simulation. Salinity variability (σ^2) is largest at zone MR and decreases toward the west for both observations and simulation. The standard

deviation of salinity errors ($S_{err[std]}$) and salinity gradient errors (dS/dl_{err}) also decrease from the east to west. A Taylor Diagram presented in Fig. 4 summarizes some of these results visually.

The variability of simulated salinity is similar to that of observed salinity (Fig. 4). With many data within the area of low RMSD and high correlation along the normalized standard deviation of 1, the model demonstrates its ability to simulate salinity reasonably well. Comparing the data distribution for all zones, there appears to be an east–west trend in the variability of the data, with zone MR the most variable and zone TX the least variable (Fig. 4). The simulated salinity in zone MR is variable due to a large salinity error near the mouth of the Mississippi River where accurate salinity simulation is known to be difficult (Hetland and DiMarco 2012); small sample size in this region may also cause the large error (Table 1). There appears to be interannual variability in the model skill (e.g., year 2000 has a poor model performance, while 2012 performs much better). This trend is associated with the Mississippi River streamflow and is addressed in Section 3.8.

Fig. 4 Taylor diagram. Two digit numbers represent the last two digits of year (e.g., 00 is 2000). REF indicates reference value (i.e., normalized standard deviation of 1). The colors represent each zone. X and Y axes are the normalized standard deviations and the arc represents correlation coefficient. The green dashed lines are root-mean-square-error (RMSE). In short, the closer the value to the REF, the better the performance of the model



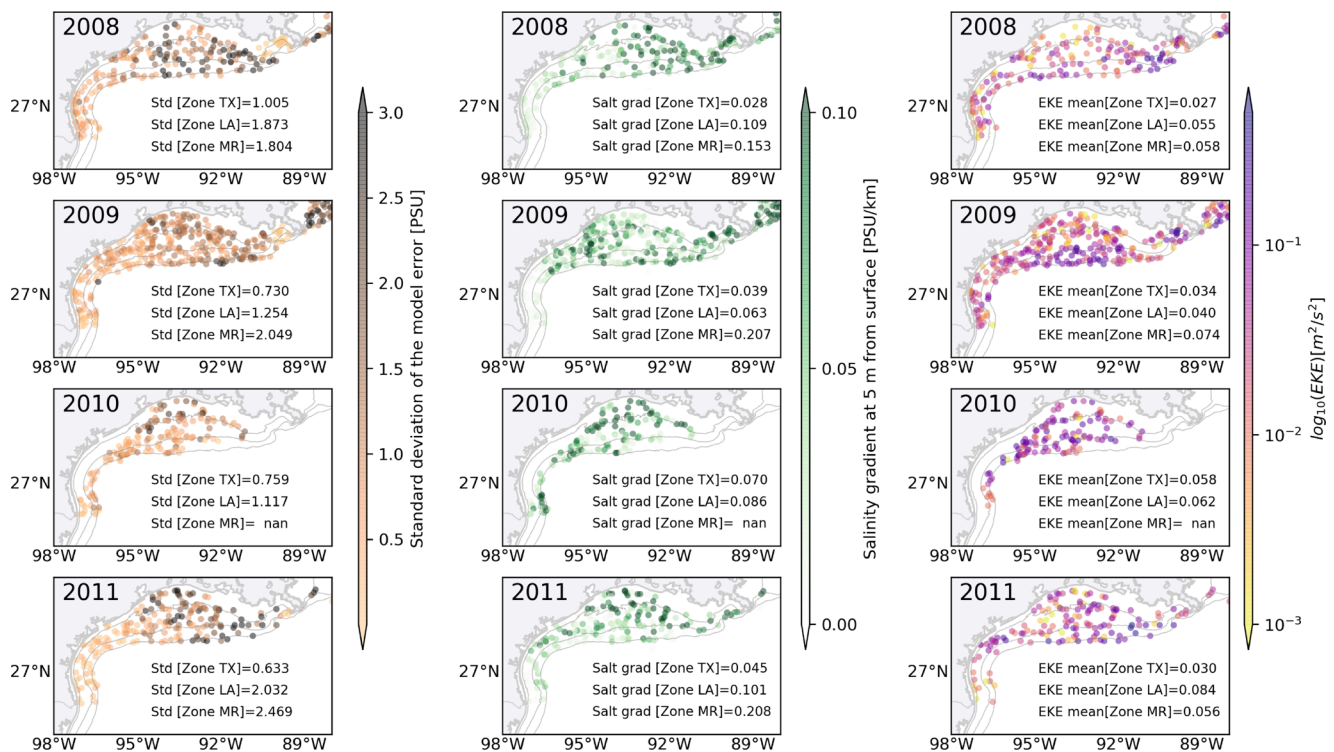


Fig. 5 Snapshots of standard deviation of model error averaged over top 20 m (left), surface salinity gradients (middle), mean surface EKE (right) from 2008 to 2011

3.6 Model error for salinity

In order to investigate the performance of salinity variability associated with frontal eddies, the standard deviation of the model error of instantaneous salinity is estimated (Fig. 5). The analysis only includes data taken in summer, when the eddies are strongest, and includes only data within the top 20 m; as a comparison, statistical parameters for both summer and fall are presented in Table 2. The regions of

larger model error are concentrated in zones LA and MR (left column in Fig. 5; also see Table 2). The large error at zone MR is believed to be associated with the details of the near-field Mississippi Delta outflow, not frontal eddies. The large error is evident in the nearshore (depth shallower than 20 m) and is scattered in the mid-/outer shelf (depth deeper than 20 m) except for the area off the southwestern Texas. For reference, the large model error at zone LA and zone MR is largely absent over the mid- and outer shelf for fall.

Table 2 Statistics of salinity errors at 5 m from surface and surface EKE for each zone. Values on top are for summer and those on bottom are for fall

Parameter	Zone	2008	2009	2010	2011
$(S_o - S_m)_{std}$	TX	0.836 0.420	0.619 0.295	0.808 0.723	0.703 0.293
	LA	1.824 0.315	1.125 0.751	1.102 0.618	1.849 0.458
	MR	2.017 0.541	1.853 0.739	NA 0.744	2.077 0.340
EKE_{mean}	TX	0.027 0.018	0.034 0.014	0.058 0.030	0.030 0.014
	LA	0.055 0.008	0.040 0.019	0.062 0.014	0.084 0.012
	MR	0.058 0.017	0.074 0.020	NA 0.011	0.056 0.007

In contrast to the nearshore water where the model error is often large (e.g., 94° W–92° W) (Table 2), the locations of the large model error in the mid- and outer shelf are not consistent among the 4-year period and the error seems to occur in random locations. The averaged error increases from the west to east (i.e., zone TX to zone MR) in summer, whereas there is not such a trend for salinity in the fall data (Table 2). There is a seasonal difference, too. The average error in summer is almost always higher than the error in fall (Table 2). When averaging the variability of the model error over the years presented in Table 2, the errors in summer are 1.87, 3.14, and 3.62 times higher than the errors in fall for zones TX, LA, and MR, respectively.

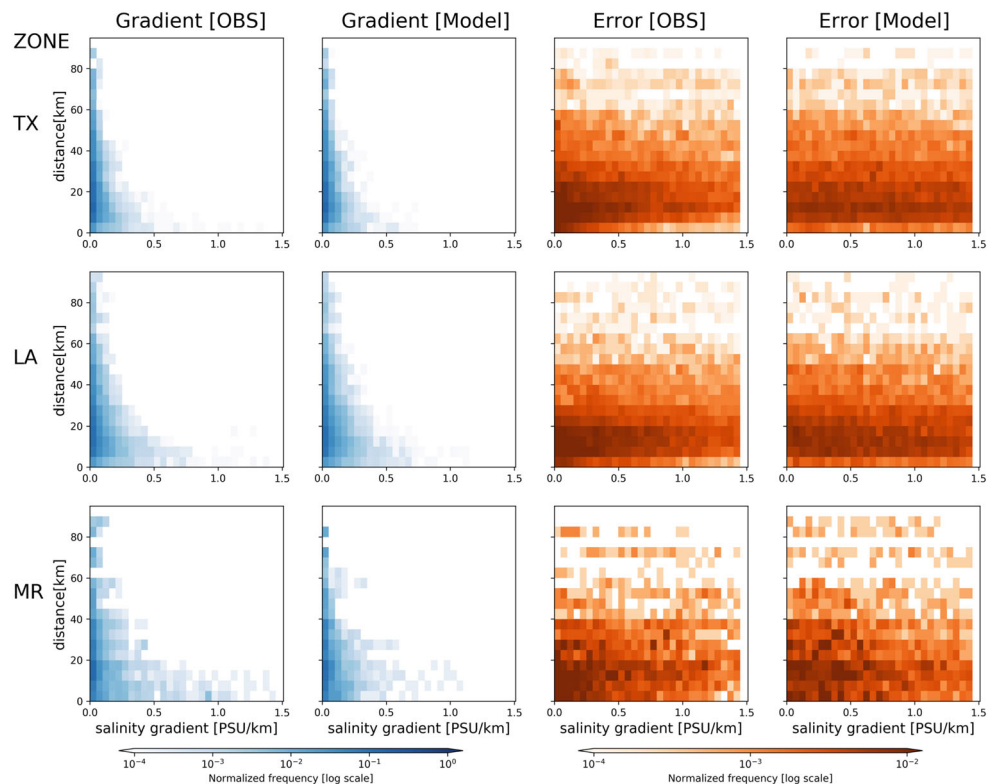
The observed salinity gradients ($\frac{dS}{dl}|_{obs}$) show a distribution similar to that of the model error (Fig. 5 middle). The distribution of observed salinity gradients and EKE is also similar. Here, EKE is estimated based on the model output, at locations extracted nearest to the hydrographic stations using the method described in Section 3.4. There is a significant statistical relationship between the model error, salinity gradient, and EKE; the model error is positively correlated with both salinity gradient and EKE ($R > 0.66$, $p < 0.027$). There is a spatial variation on salinity gradient and EKE. They both increase from the west to east with the largest value on zone MS on average except for 2011 when the number of sampling stations is much less than that of the other years (Fig. 5 right and Table 2). The high salinity gradient and EKE at zone MR are possibly influenced

by a quasi-stationary anti-cyclonic eddy west of the mouth of the Mississippi River (Wang and Justić 2009) and the eddies on the Louisiana Bight east of the river mouth moving eastward. Both are not considered as frontal eddies on the TXLA shelf.

3.7 Statistical representation of frontal eddies

The model results indicate that the model is able to replicate mean salinity structure both horizontally and vertically. However, the model error is known to be largest in the region where frontal eddies are actively present (Marta-Almeida et al. 2013). As illustrated in the normalized vorticity presented in Fig. 1a, the eddies are characterized by anti-cyclonic (clockwise) vorticity in their core, with strong ($\zeta/f > 1$) cyclonic vorticity along the eddy perimeters. These cyclonic vorticity contributes to a shoreward salt flux—low-salinity water to the offshore and high-salinity water to the inshore—in addition to the summertime offshore Ekman transport. Though the model appears to show a poor performance in predicting salinity associated with frontal eddies precisely since the locations of the eddies are random, it is not clear if the scale of frontal eddies can be replicated by the model. In particular, is the model able to reproduce the strong salinity gradients associated with the high vorticity along the eddy perimeters? In order to investigate this, salinity gradients in the form of histogram are calculated based on the hydrographic ($\frac{dS}{dl}|_{obs}$) and the model data

Fig. 6 Two-dimensional histograms. (1st column from left) Observed salinity gradient, (2nd column) simulated salinity gradient, (3rd column) error based on observation, (4th column) error based on model. The observational error is random error with normal distribution of salinity gradients. The model error is standard deviation of salinity gradient error for the time closest to the observations and adjacent hours (-1 and +1 h). See the text for details



($\frac{dS}{dl}|_{sim_sampled}$) sampled in a similar way; the results are presented in Fig. 6 where only summer data are presented.

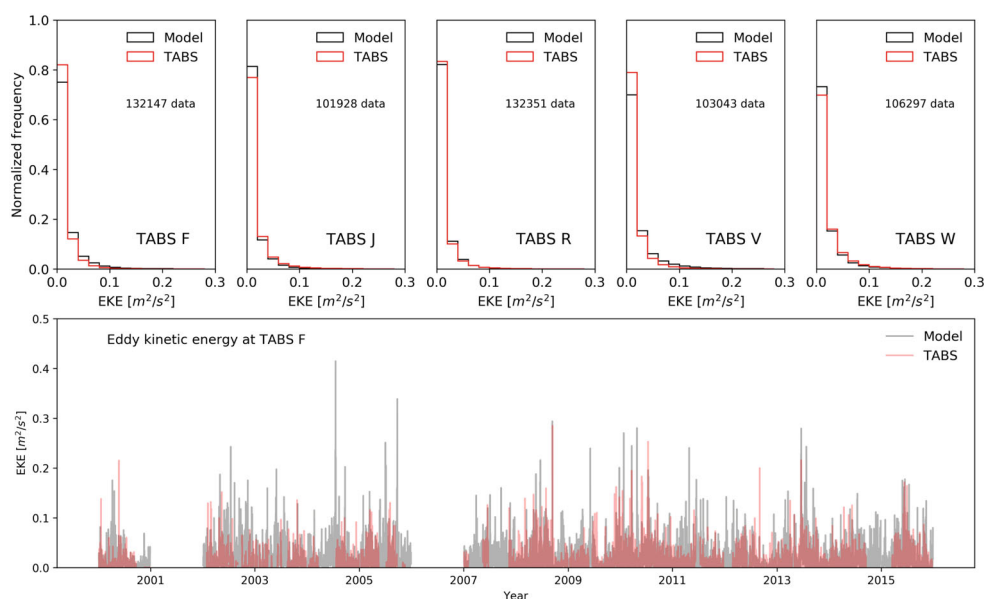
The low gradients may occur in both close and distant stations, since it is possible that many fluctuations in salinity occur between the stations, but high gradients only occur in stations with a short separation distance. Of course, the strongest gradients that can be observed or simulated depend on the sampling pattern and model resolution, respectively. The observed salinity gradients are indeed strongest at stations near each other, and the model reproduces the pattern. In particular, the agreement in salinity gradients between the observation and the model is good for both low and high salinity gradients across all separation distances. We identify that salinity gradient at the edges of each eddy front appears to match with $0.05 \text{ g}\cdot\text{kg}^{-1}\cdot\text{km}^{-1}$ or higher depending on the size and intensity of the eddies, and therefore use the value as a reference value. The frequency of high salinity gradients increases from zone TX to zone MR (Fig. 6). For fall, the agreement between the model and observations is similar to that for summer, but the frequency of the high salinity gradients is much lower than that of the summer counterpart (figure not shown). The strong gradients ($>0.05 \text{ g}\cdot\text{kg}^{-1}\cdot\text{km}^{-1}$) for shorter distance (0–30 km) are believed to be associated with the buoyancy-driven coastal current as well as frontal eddies. How much of the high salinity gradients are associated with frontal eddies varies for each zone. Since the buoyant coastal current seems to be dominated for the region with depths of 20 m or shallower, we extracted the salinity gradients that are higher than $0.05 \text{ g}\cdot\text{kg}^{-1}\cdot\text{km}^{-1}$ and are deeper than 20 m for the separation distances between 0 and 30 km (i.e.,

submesoscale). If we can assume that high salinity gradients located at depths deeper than 20 m are primarily associated with frontal eddies, a large portion of the high salinity gradients (i.e., 83, 74, and 92% of the data samples for zones TX, LA, and MR, respectively) would be associated with frontal eddies on the shelf. The percentage of zone MR does not necessarily show an influence of salinity gradients in frontal eddies because a large portion of high salinity gradients are not thought to be associated with frontal eddies on the TXLA shelf as previously mentioned. The frequency of the model error in salinity gradient (fourth column in Fig. 6) looks very similar to that of the random noises added to the observations (third column in Fig. 6) in particular the high-frequency band between 0 and 30 km. Both the model error and the noises are scattered across salinity gradients and separation distances. As for the fall data, a similar pattern is given with smaller salinity gradients at the low separation distances.

The EKE estimated from TABS stations as presented in Fig. 7 shows a similar result. A time series of EKE from both the observations and the modeling at TABS F (Fig. 7 bottom) shows a similar variability though they occasionally disagree in magnitude. The histograms demonstrate similar shapes for all stations presented in Fig. 7 (Fig. 7 top).

Variability of the model performance is further analyzed by comparing monthly salinity gradients averaged over June through August for 16 years and the model error similar to what is presented in Fig. 4. Note that for this particular purpose, we extracted the data for the shaded region presented in Fig. 3 where frontal eddies are most active. As shown in Fig. 12, salinity gradients have a

Fig. 7 (Top) Histograms of EKE for different TABS stations. (Bottom) Time series of EKE for model (gray) and observation (red) at TABS F



positive correlation with the RMS model error of salinity and the correlation is strong.

3.8 Seasonal and inter-annual variability of frontal eddies

All variables in Fig. 8 indicate high-frequency and seasonal variability. The high-frequency spikes of wind stress in winter are associated with the passage of cold fronts. Other spikes are tied with other meteorological events (e.g., tropical cyclones). A portion of the high-frequency variability of eddy parameters corresponds to the spikes. Although the alias EKE signals are reduced, they are not fully eliminated. These high-frequency spikes are not believed to be associated with baroclinic instability.

The general seasonal trends are such that (i) salinity gradients, $\frac{dS}{dt}|_{sim}$, and EKE are maximum in summer and minimum in winter, (ii) streamflow is maximum in spring and minimum in summer, and (iii) wind stress is maximum in winter and minimum in summer. Such seasonal variability is clearly evident in Fig. 9, which

shows time series of annual averaged and monthly averaged parameters presented in Fig. 8. When comparing annual streamflow (Fig. 9a) to other eddy variables (EKE and salinity gradient) (Fig. 9b), it is visually evident that when the annual streamflow is lower (e.g., years 2000 and 2006), the eddy variables also show lower values for the years. Thus, there should be some correlations between them.

In order to compare the relationship of all parameters, we create a scattered plot matrix based on monthly averaged data of 23 years (Fig. 10). In Fig. 10, each consists of a pair of variables and the diagonal plots are kernel density estimator (KDE). It is noted that the relationship between each pair may not necessarily be linear. We have included the linear correlation coefficient, R , merely to show how good both data fit. It demonstrates that the correlation between EKE and salinity gradients is very strong ($R = 0.9$, $p \ll 0.05$). It also shows that streamflow and salinity gradients have a positive correlation with the correlation coefficient of 0.7 ($p \ll 0.05$). It is noted that since it takes certain time for freshwater to be covered onto the

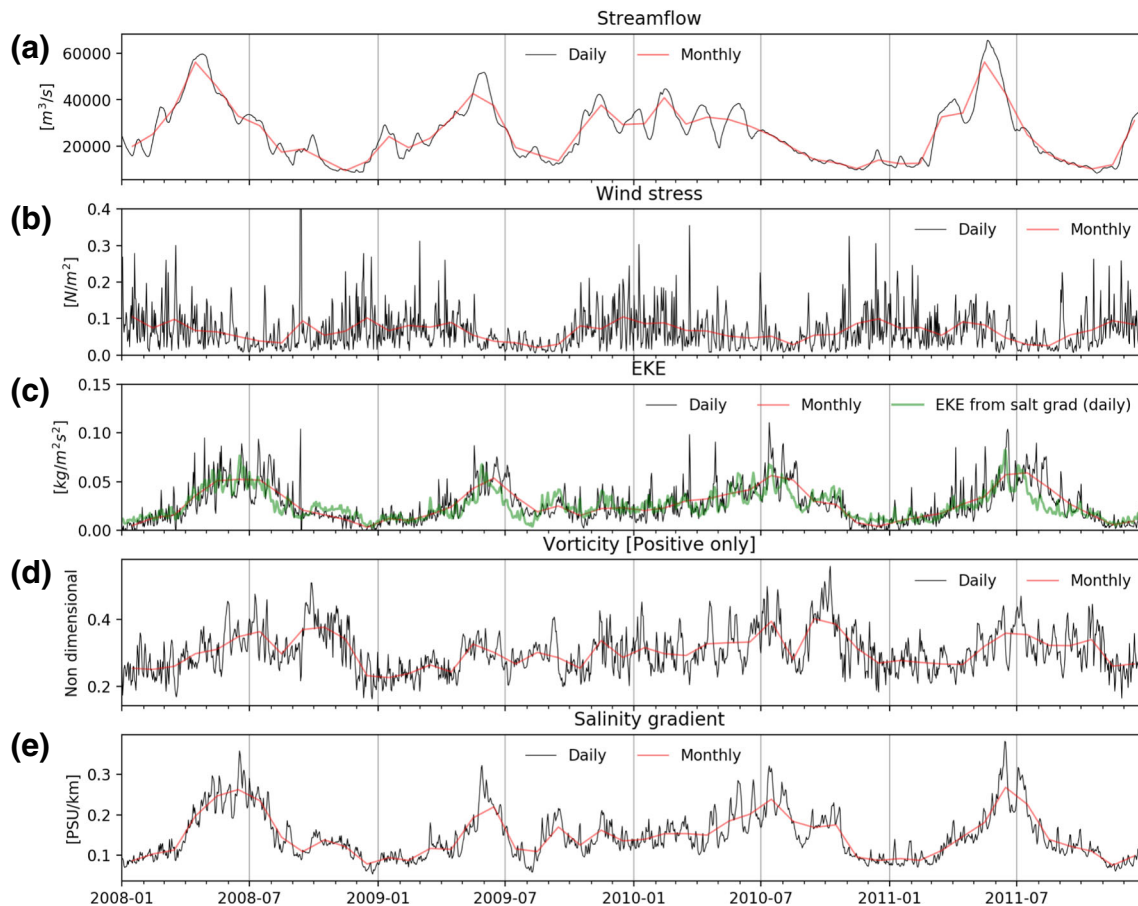


Fig. 8 Time series of daily-averaged (a) streamflow, (b) wind stress, (c) EKE, (d) positive vorticity, and (e) salinity gradient. Red lines are monthly averaged values. The green line represents EKE estimated

from salinity gradient using Eq. 4. The data presented in the figure are from the model outputs except for the streamflow

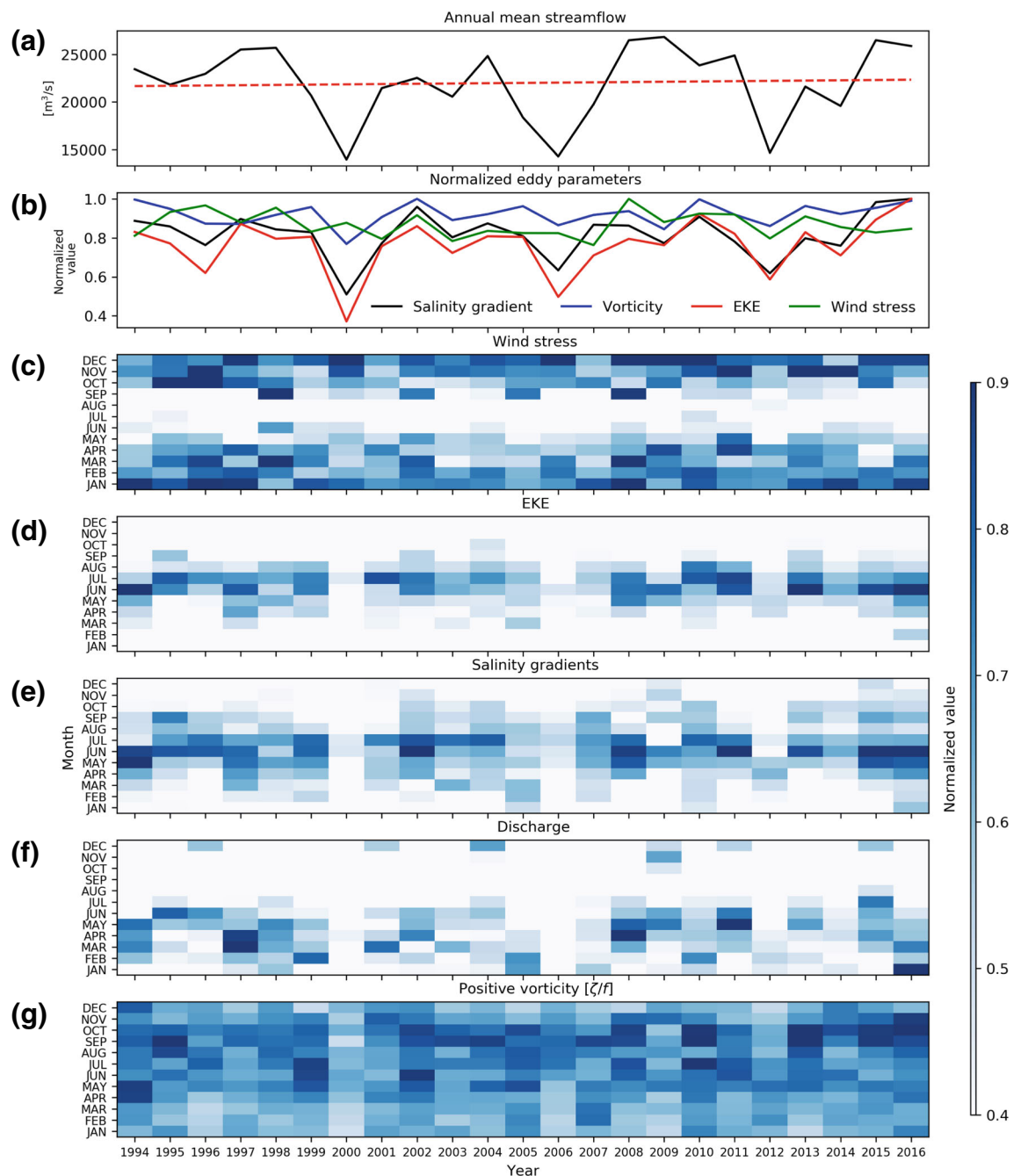


Fig. 9 Time series of (a) annual mean streamflow, (b) eddy variables, (c) wind stress, (d) EKE, (e) salinity gradient, (f) streamflow, and (g) positive vorticity. (c)–(g) show shaded plots and the values are normalized so that the maximum value is 1

shelf, there is a time lag between streamflow and eddy variables. The lag is estimated to be 2 months. Figure 10 also shows that wind stress and salinity gradients have an inverse correlation ($R = -0.54$, $p \ll 0.05$). It is interesting to see that the normalized vorticity has a poor correlation with EKE ($R = 0.45$, $p \ll 0.05$) compared with the relationship between EKE and salinity gradients ($R = 0.9$,

$p \ll 0.05$). When looking at Fig. 9g, the vorticity has a second maximum in fall; while, EKE and salinity gradients do not. Such second maximum does exist for kinetic energy (KE) (figure not shown).

The finding of a strong correlation between EKE and salinity gradients suggests a possibility to estimate EKE from salinity gradients ($\frac{dS}{dt}|_{sim}$) and eliminate the alias

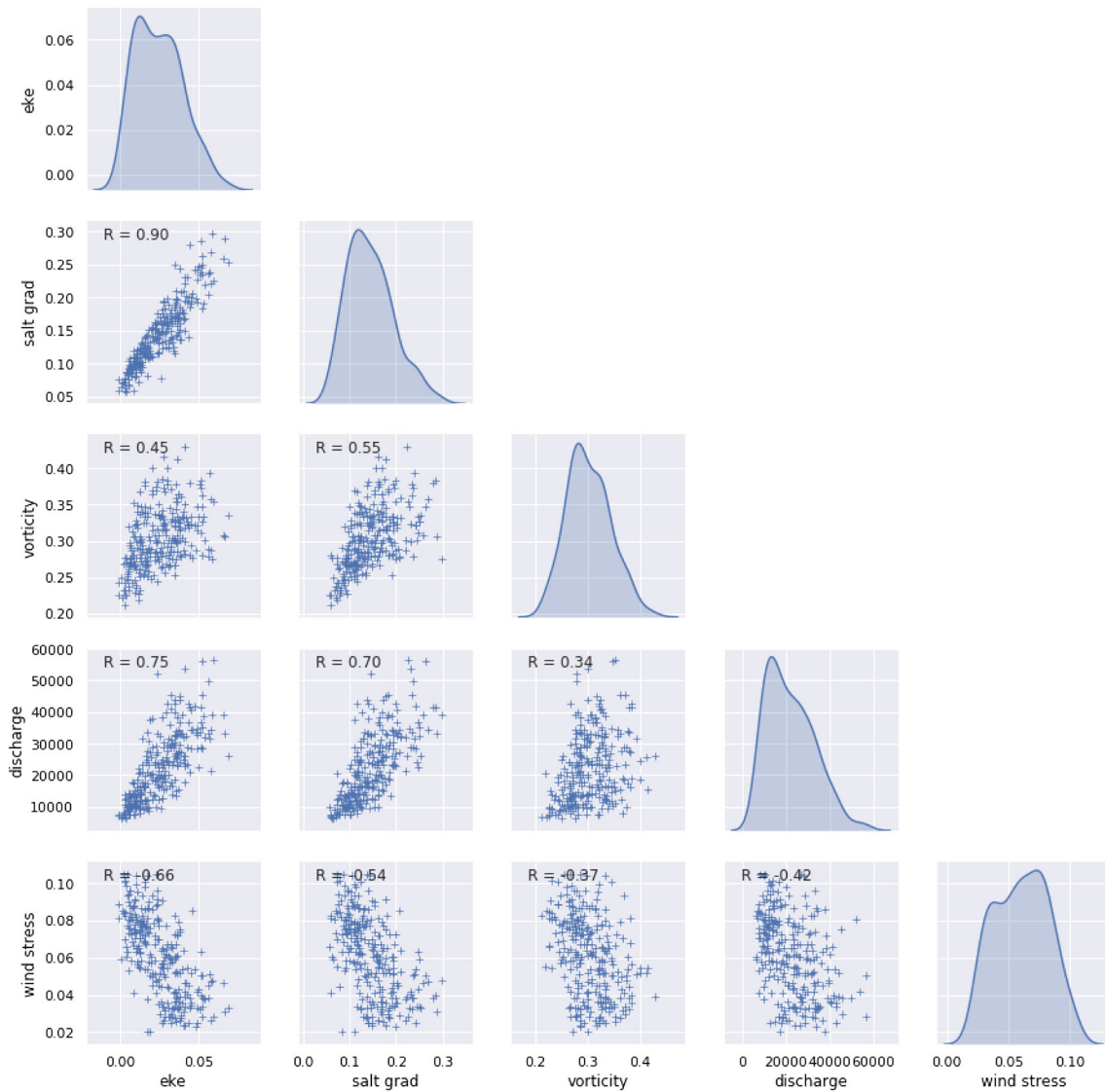


Fig. 10 A scattered plot matrix. Diagonal plots are kernel density estimate (KDE). All pairs show p value smaller than at least 10^{-8}

signals of EKE. However, a question remains as to whether the relationship between EKE and salinity gradients is linear or non-linear. Fox-Kemper et al. (2008) developed a parameterization of restratification for mixed layer eddies (MLE). Using the scaling they developed, EKE can be related to the square salinity gradients as described below.

$$\begin{aligned} \frac{\partial EKE}{\partial t} &\approx \langle w'b' \rangle \\ &= |\nabla_h b|^2 \cdot MLD^2 / f \\ &\sim \left(\frac{dS}{dt} \Big|_{sim} \right)^2 \cdot MLD^2 \end{aligned} \quad (4)$$

Where $\langle w'b' \rangle$ is vertical buoyancy flux averaged over the selected region (see Fig. 3 for the region), $\nabla_h b$ is horizontal buoyancy gradient, and MLD is surface mixed layer depth. Using the Eq. 4 and given that MLD has small variability

on the shelf, EKE can be estimated from salinity gradient by:

$$\begin{aligned} EKE &\sim \int \left\{ \left(\frac{dS}{dt} \Big|_{sim} \right)^2 \cdot MLD^2 \right\} dt \\ &= \left(\frac{dS}{dt} \Big|_{sim} \right)^2 \cdot MLD^2 \cdot T + C \\ &\simeq \alpha \cdot \left(\frac{dS}{dt} \Big|_{sim} \right)^2 + \beta \end{aligned} \quad (5)$$

where α (in m^4/s^2) and β (in m^2/s^2) are constants, T is a representative time scale, and C is a constant. Here, β needs to be greater than or equal to 0. The result of our preliminary analysis indicates that the relationship between EKE and $\langle w'b' \rangle$ seems to be linear (Fig. 11 top) and the relationship between EKE and salinity gradient appears to be non-linear

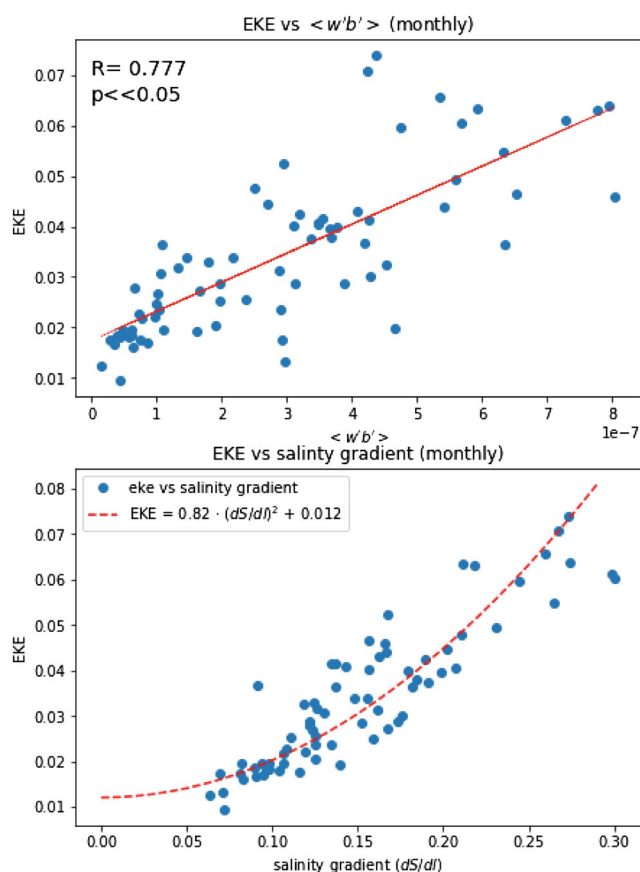


Fig. 11 Scattered plots of (top) $\langle w'b' \rangle$ and EKE and (bottom) EKE and salinity gradients. The curved line on the bottom figure was drawn with $\alpha=0.82$, $\beta=0.012$, which were determined based on a curve fitting analysis

(i.e., square of salinity gradients) (Fig. 11 bottom). The EKE estimated from Eq. 5 eliminates high-frequency EKE spikes in winter (green line in Fig. 8c).

At inter-annual scales, salinity gradients in summer positively correlate with streamflow (Fig. 12 middle). There is an increase in trend of streamflow and EKE. The streamflow across the Texas and Louisiana Rivers, over 90% of which is from the Mississippi and Atchafalaya Rivers, increases over the period of 23 years at the rate of $30.35 \text{ m}^3 \cdot \text{s}^{-1} \cdot \text{year}^{-1}$ (Fig. 9a, Table 3). While wind stress shows a slight decrease over the same period ($-8.3 \times 10^{-5} \text{ N} \cdot \text{m}^{-2} \cdot \text{year}^{-1}$) (Table 3). EKE and salinity gradients also show an increase in trend over the same period. During the period, the average of EKE and salinity gradients increases by 13 and 4%, respectively. The increase in trend of EKE is consistent with an increase in trend of streamflow and a decreasing trend of wind stress. The finding is in line with the seasonal variability; however, the driving mechanisms are likely different (seasonal vs inter-annual).

4 Discussion

The focus of this paper is to investigate reproducibility of salinity fields caused by shelf frontal eddies and their variability. We first evaluated predicted salinity against observations with the densely sampled CTD data from SEAMAP collected for the period between 2000 and 2015. There are very few papers evaluating ocean eddies with the scale smaller than mesoscale due to scarcity of available field measurements, and consequently, most of relevant studies focus on idealized and semi-idealized simulations. Also, to the author's knowledge, virtually no scientific papers have discussed seasonal and inter-annual variability of submesoscale eddies for the shelf seas under the influence of freshwater. Therefore, this paper provides a unique opportunity to address the validity of submesoscale eddy simulation and the variability of the eddies of the shelf seas.

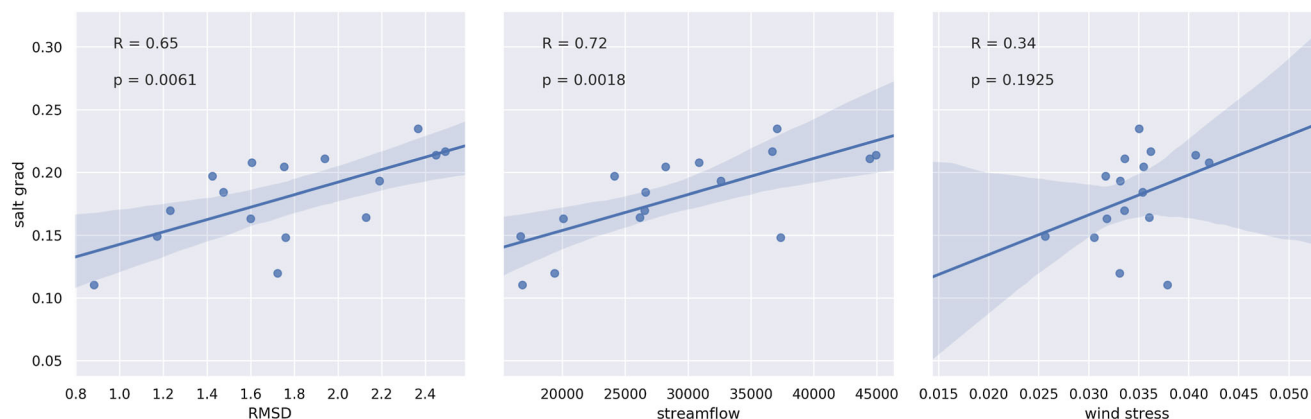


Fig. 12 Scattered plots of (left) RMSD and salinity gradient, (middle) streamflow and salinity gradient, and (right) wind stress and salinity gradient. The data are for summer only

Table 3 Linear trend of total annual average streamflow ($\text{m}^3/\text{s}/\text{year}$), salinity gradients ($\text{g}/\text{kg}/\text{km}/\text{year}$), EKE (J/year), and wind stress ($\text{N}/\text{m}^2/\text{year}$). Note that the mean streamflow for summer is the average between April and June

Means	Streamflow	EKE	Salinity gradient	Wind stress
Annual mean	30.35	0.00014	0.00024	-8.3×10^{-5}
Summer mean (Jun–Aug)	−29.02	0.00018	−0.00024	-4.43×10^{-5}

4.1 Is a non-assimilative ocean model able to replicate submesoscale eddies?

The analysis of model outputs demonstrates robust performance in salinity simulation. The large model error is in line with the locations where EKE and salinity gradient are high; thus, it should be caused largely by mismatches on the eddy locations. This view is supported by the strong correlation between model error (RMSD) and salinity gradients ($R = 0.65$, $p = 0.0061$). The model performance varies spatially and temporarily. It shows that EKE and salinity gradients are higher in summer than non-summer seasons. The spatial variation is revealed for model errors, too. It shows the east–west trends on the model error with the east (i.e., nearfield) the lowest performance. The highest EKE at zone MR is possibly influenced by a quasi-stationary anti-cyclonic eddy west of the mouth of the Mississippi River (Wang and Justic, 2009) and the eddies on the Louisiana Bight east of the river mouth moving eastward. Both are not considered as frontal eddies on the TXLA shelf; thus, it is considered that zone LA be the eddy hotspot over the TXLA shelf.

The model ability to statistically replicate frontal eddies through salinity gradients demonstrates its robustness on frontal eddy simulation. Since salinity gradients are correlated with EKE (Fig. 10), the non-assimilative model demonstrates its ability to reproduce the cumulative effects of frontal eddies for the Texas-Louisiana shelf. In other words, although the model is not able to replicate the locations of the eddies precisely, the model should be able to replicate the cumulative eddy kinetic energy that is equivalent to the observations. This view is corroborated by the comparison in EKE between the model and the fixed-point measurements at various surface mooring locations (Fig. 7).

4.2 What influences the variability of frontal eddies on the shelf?

Analysis of eddy variability using the 23-year hindcast data demonstrates seasonal and inter-annual variability. At seasonal scale, the eddy variables, namely, EKE, and salinity gradients, are positively correlated with streamflow and negatively with wind stress. An increase in EKE and salinity gradients with an increase in streamflow is not necessarily surprising. Density fronts are a necessary

condition for baroclinic instability and the resulting eddies. The freshwater retention period for the TXLA shelf is approximately 3–6 months (Zhang et al. 2012); thus, the high volume of freshwater discharged from the rivers in spring should stay on the shelf in the upcoming summer. Such a high volume of freshwater creates sharper density fronts on the shelf in summer, which in turn increases lateral salinity gradients (and also lateral buoyancy gradients). Since salinity gradients are positively correlated with EKE, the increase in freshwater volume elevates EKE.

The inverse correlation between wind stress and EKE appears to be tied with transient wind events such as winter storms. When the fronts exist, as they are physically unstable, the fronts slump and the gravitational adjustment leads to lateral buoyancy transport, and consequently, re-establishment of vertical stratification. The decrease in salinity gradients during the passage of the fronts supports our interpretation (figure not shown). However, pluses of strong winds such as those during winter storms can induce vertical mixing, destroy the stratification, and eventually, weaken lateral density gradients in a matter of days (e.g., Simpson and Sharples 2012). In addition, in winter net surface heat flux is negative (i.e., the ocean loses heat to the atmosphere), triggering convective mixing (Fig. 13b). The convective mixing in the surface mixed layer may also contribute to the weakening of plume fronts as well. N^2 (buoyancy frequency squared), which represents stability of water column, shows seasonal variability with the lowest N^2 in winter (Fig. 13d); thus, lateral density gradients tend to weaken in winter. Reduction in lateral density gradients weakens instabilities, and consequently, leads to lower EKE. Hetland (2017) showed based on idealized simulation that instabilities grow 3 days after the model initialization and last for 6 days when EKE reaches a saturation state. For the northern Gulf of Mexico, cold fronts pass over the region once every 3 to 10 days (DiMego et al. 1976). Wind-induced mixing in the surface mixed layer during a winter storm can reduce lateral density gradients (and consequently, weaken instabilities) and it may take 3–9 days for the instabilities to be fully re-developed. However, another cold front is expected to pass over the region before sharp fronts are fully re-established, inhibiting the instabilities to grow in winter. On the other hand, in summer, in the absence of tropical cyclones, there are few storms. As a result, the mechanical energy to weaken the fronts is virtually non-existent. Coinciding that the volume of freshwater on the

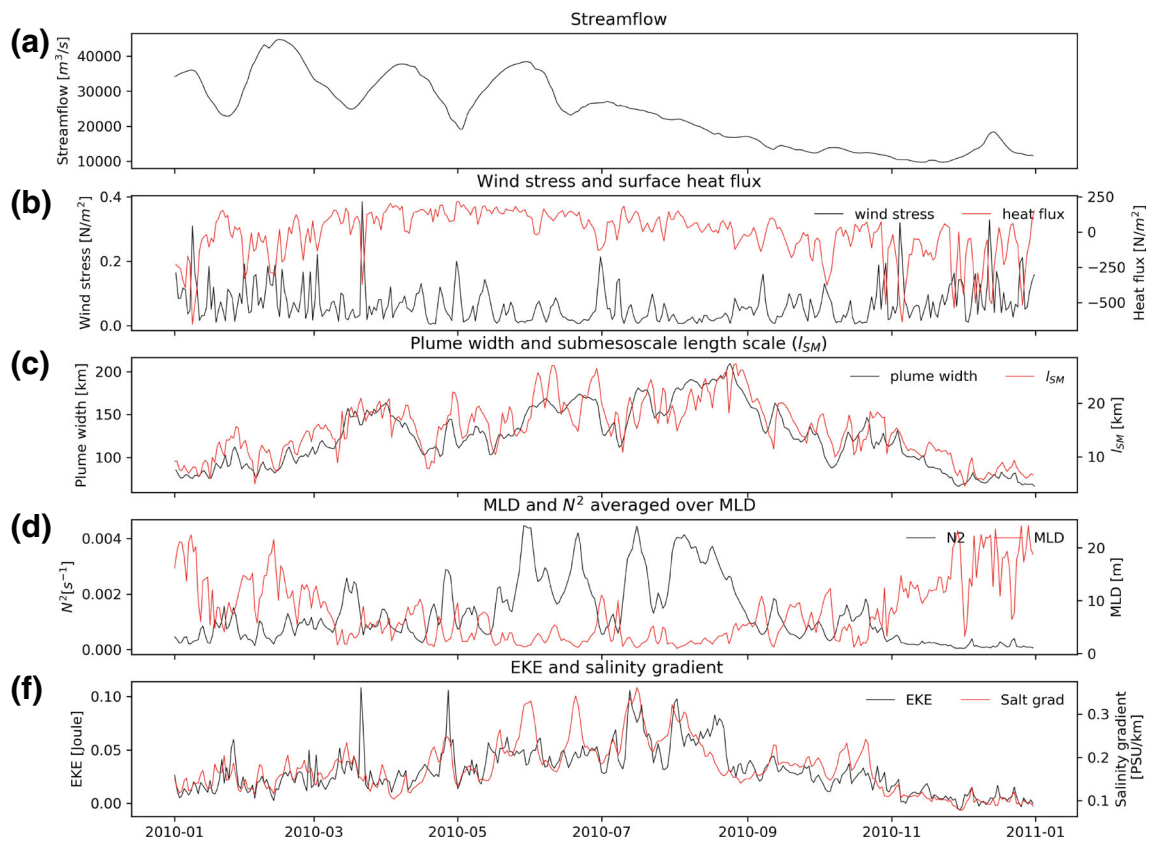


Fig. 13 Time series of daily averaged (a) streamflow, (b) wind stress and surface heat flux, (c) plume width and submesoscale length scale, (d) MLD and N^2 averaged over MLD, and (e) EKE and salinity gradient

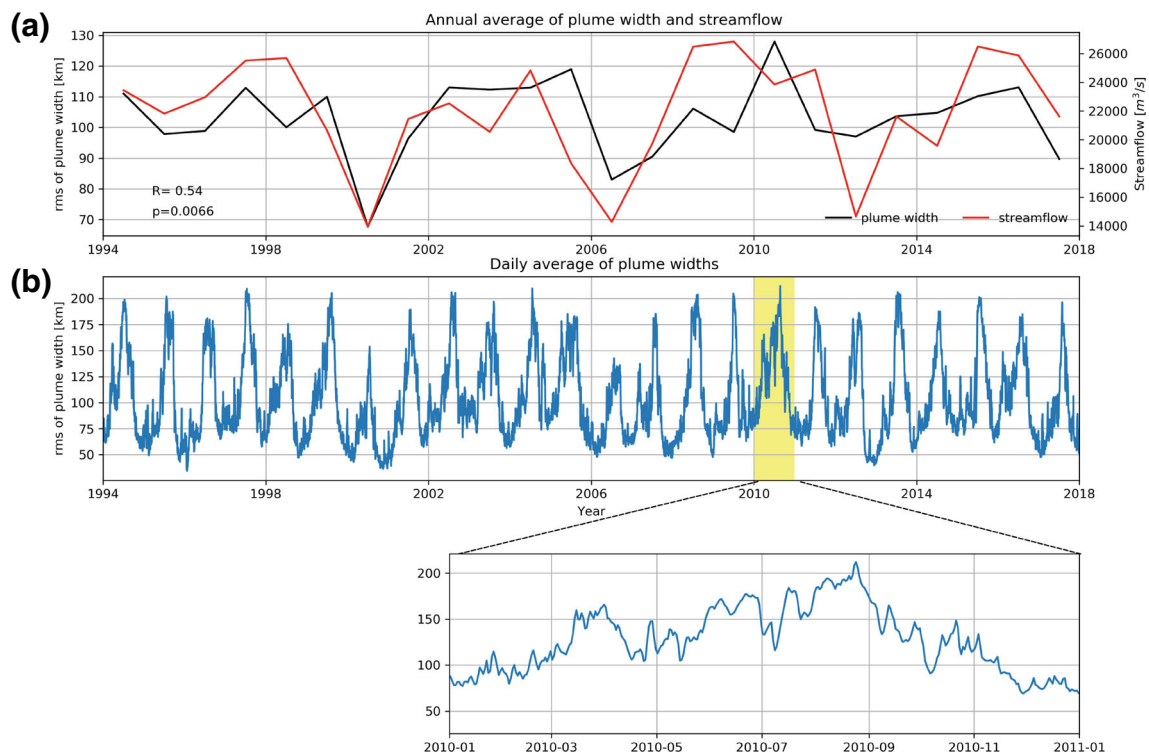


Fig. 14 Time series of a annual mean streamflow and plume width and b daily averaged plume width. The embedded figure shows daily plume width for 2010

shelf is the highest in spring to summer and that winds are upwelling-favorable, which helps broaden the plume (see Section 4.3), summer is in favor of the instabilities to develop.

At inter-annual scales, the positive correlation between streamflow and EKE suggests a potential link between submesoscale eddies and climate variability. Variability of streamflow is known to link climate variability. For example, there are reports that extreme streamflow in the Mississippi River links to ENSO (El Niño Southern Oscillation) and AMO (Atlantic Multidecadal Oscillation) (Munoz and Dee 2017; Munoz et al. 2018). The river has also been highly engineered, and thus, the streamflow conditions have changed naturally and artificially (e.g., sediment and freshwater diversion) (Munoz et al. 2018). Consequently, submesoscale eddies can be remotely influenced by such natural and anthropogenic variability. The remote influence in submesoscale eddies is intriguing. However, establishing the quantitative relationship between river climatology and climate variability is not easy. We compared streamflow data at Tarbert Landing and a few climate indicators, namely Southern Oscillation Index (SOI), and AMO index (Supplemental Figure S3). The relationship between ENSO, AMO, and freshwater volume is not clearly evident in our preliminary analysis despite the known links. Despite the difficulty, it is suggested that if streamflow varies with climate variability and also a changing climate, submesoscale eddies are also expected to be influenced by the long-term variability as well because of the robust relationship between streamflow and frontal eddies.

4.3 What mechanisms control frontal eddy variability?

Over a broad, low-energy shelf off Texas and Louisiana, submesoscale frontal eddies are widespread in summer. It is reported that the development of submesoscale instabilities links to two key parameters in open water: lateral buoyancy gradients and mixed layer depth (MLD) (e.g., Capet et al. 2008a; Mensa et al. 2013). Since fronts are ubiquitous regardless of seasons in open water in particular near boundary currents, seasonality of MLD seems to play a key role in variability of submesoscale instabilities. In winter, deep mixed layer driven by stronger wind-induced mixing provides reservoirs of available potential energy (APE), which allows re-circulation cell along the perimeter of the eddies to develop by converting APE to EKE, whereas in summer shallow surface mixed layer driven by intense solar heating prevents submesoscale circulation cell from developing, inhibiting instabilities (Mensa et al. 2013). Thus, submesoscale instabilities are known to be most active in winter in open water (e.g., Capet et al. 2008b). Over the shelf seas, the driving mechanisms are somewhat

similar, but there are at least three distinctions from those for open water: shallow bathymetry (and consequently shallow MLD regardless of seasons), land boundary, and terrestrial influences. In particular, freshwater influx changes lateral buoyancy gradients (∇b_H) and plays a role in mixed layer instabilities (e.g., McWilliams 2016). As discussed by Hetland (2017), submesoscale instabilities for a broad shelf under the influence of freshwater are thought to be driven by a couple of factors: lateral density gradients and bottom slopes. Hetland (2017) mentioned that plume width may be a critical factor that baroclinic instability develops as frontal eddies do not fit for narrow plume. We estimate the plume width that is regarded as the distance between the coastline and the isohaline of 33 g·kg⁻¹ (Figs. 13c and 14). The result indicates clear seasonal variability with the narrowest width in winter (~50 km) and the widest in summer (~200 km) (Fig. 14).

Hetland (2017) also mentioned that if the plume width is smaller than five times the baroclinic deformation radius ($W < 5R_d$ where W is plume width and R_d is the deformation radius), the instabilities may not form. For submesoscale eddies, R_d is equivalent to $R_{d[ML]}$, which is mixed layer deformation radius ($R_{d[ML]} = \sqrt{MLD * g * \Delta\rho / \rho_0 / f}$ (McWilliams 2016; Özgökmen et al. 2011)). Mensa et al. (2013) calculated submesoscale length scale, l_{SM} ($= 5R_{d[ML]}$) based on Eldevik and Dysthe (2002). l_{SM} , which represents the most unstable wavelength of submesoscale instabilities, is considered to be the upper end of submesoscale eddy diameter, D ($R_{d[ML[min]]} < D < l_{SM[max]}$) (Eldevik and Dysthe 2002). l_{SM} ranges from 5 to 30 km with its maximum value in summer for the TXLA shelf (Fig. 13c) and it roughly matches the spatial scale of submesoscale eddies on the shelf (Fig. 1a). If the relationship proposed by Hetland (2017) holds for the realistic situation, submesoscale instabilities indeed can be suppressed in winter, but not in summer for the TXLA shelf. The finding is in clear contrast to other similar studies. Interestingly, the variability of the length scale l_{SM} is remarkably similar to that of the plume width (Fig. 13c), corroborating the importance of the plume width on submesoscale instabilities. We also compare salinity gradients, plume width, and submesoscale length scale between the shipboard observations and the modeling and the agreement is good, demonstrating the validity of our submesoscale simulation (see Supplemental Figure S2).

The relationship between EKE and salinity gradients as well as EKE and $\langle w'b' \rangle$ has been identified. Although a further analysis will be needed in a separate paper, the nonlinear relationship between EKE and salinity gradients suggests that the scaling proposed by Fox-Kemper et al. (2008) is valid for our simulation, which further suggests that the shelf frontal eddies driven primarily by large influx of freshwater from rivers have a similar dynamical

characteristic to submesoscale eddies in open water. In addition, given the validity of the relationship between EKE and $\langle w'b' \rangle$ in our submesoscale eddy simulation, it appears that the model replicates mixed layer instabilities at submesoscale in a realistic setting (Capet et al. 2008a; Fox-Kemper et al. 2008; Mensa et al. 2013).

The annual average plume width indicates that the plume width is positively correlated with annual streamflow (Fig. 14); it also appears that the daily plume width correlates with streamflow with a time lag of roughly 2 months (Fig. 13a and c). Therefore, it is suggested that the volume of freshwater plays a key role in plume width, and consequently, the formation and development of submesoscale instabilities. In addition, the plume width and wind stress are *inversely* correlated for both high-frequency band (a day to weeks) and seasonal band (Fig. 13b, c and Figure S4). The influence of plume width on high-frequency wind stress seems to be secondary and the seasonal variability is dominated (Fig. 13c). Such seasonal variability is more clearly visible for monthly averaged data for which wind stress and plume width have a clear inverse correlation, suggesting that winds contribute negatively to the plume width (Supplemental Figure S4). The finding is in contrast to what Hetland (2017) suggested that wind mixing helps broaden the plume in winter, which may be a precondition for submesoscale instabilities in summer. For the variability of submesoscale frontal eddies, we conclude that the volume of freshwater that flows into the shelf and wind stress along with low-energy shelf environment, play the leading role in controlling the variability.

5 Conclusions

Shelf frontal eddies are an important feature along the plume fronts in coastal waters with significant implications for hydrodynamic and biogeochemical processes. However, predicting frontal eddies poses a serious challenge given the fact that the underlying process, baroclinic instability, is chaotic in nature and data assimilation is not yet a feasible option. We examined whether a non-assimilative ocean model is able to reproduce the variability and characteristics of shelf frontal eddies generated in the plume fronts of two large rivers connected to the Texas-Louisiana shelf.

How well the model is able to replicate frontal eddies significantly depends on the accuracy of salinity (or density) simulation. The model shows its ability to replicate horizontal and vertical mean structures and variability of salinity. The high model error coincided with high EKE and salinity gradients demonstrates that the mismatches in the locations of frontal eddies play a primary role in the model error, whereas the statistical analysis of salinity gradients indicates that the model is able to statistically represent

frontal eddies, which is regarded as that the cumulative eddy kinetic energy on the shelf is considered to be equivalent to the observed counterparts.

Our analysis reveals temporal and spatial variability of shelf frontal eddies at a range of different scales. The frontal eddies are strongest in the east close to the freshwater sources, and are weakest in the west furthest from the sources. Frontal eddies have seasonal and inter-annual variability. EKE is strongest in summer and weakest in winter. In winter, stronger winds act to suppress frontal eddies, whereas, in summer in the absence of storms and the resultant mechanical energy to destroy the fronts, frontal eddies are prevalent. At inter-annual scales, variability of eddies corresponds to the volume of freshwater from the rivers suggesting their potential links to climate variability and anthropogenic climate change.

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References

- Boccaletti G, Ferrari R, Fox-Kemper B (2007) Mixed layer instabilities and restratification. *J Phys Oceanogr* 37(9):2228–2250. <https://doi.org/10.1175/JPO3101.1>
- Capet X, McWilliams JC, Molemaker MJ, Shchepetkin AF (2008a) Mesoscale to submesoscale transition in the California current system. Part I: flow structure, eddy flux, and observational tests. *J Phys Oceanogr* 38(1):29–43. <https://doi.org/10.1175/2007JPO3671.1>
- Capet X, McWilliams JC, Molemaker MJ, Shchepetkin aF (2008b) Mesoscale to submesoscale transition in the California current system. Part III: energy balance and flux. *J Phys Oceanogr* 38(1):29–43. <https://doi.org/10.1175/2007JPO3671.1>
- Chelton DB, Schlax MG, Samelson RM, Ra deSzoek (2007) Global observations of large oceanic eddies. *Geophys Res Lett* 34(15):1–5. <https://doi.org/10.1029/2007GL030812>
- Chen C, Reid R, Nowlin W Jr (1996) Near-inertial oscillations over the Texas-Louisiana shelf. *J Geophys Res* 101(C2):3509–3524. <http://www.agu.org/pubs/crossref/1996/95JC03395.shtml>
- Cochrane JD, Kelly FJ (1986) Low-Frequency Circulation on the Texas-Louisiana Continental Shelf. *J Geophys Res* 91(C9):10645–10659. <https://doi.org/10.1029/JC091iC09p10645>, <http://www.agu.org/pubs/crossref/1986/JC091iC09p10645.shtml>
- Dee DP, Uppala SM, Simmons aJ, Berrisford P, Poli P, Kobayashi S, Andrae U, Ma Balmaseda, Balsamo G, Bauer P, Bechtold

- P, Beljaars aCM, van de Berg L, Bidlot J, Bormann N, Delsol C, Dragani R, Fuentes M, Geer aJ, Haimberger L, Healy SB, Hersbach H, Hólm EV, Isaksen L, Kållberg P, Köhler M, Matricardi M, McNally aP, Monge-Sanz BM, Morcrette JJ, Park BK, Peubey C, de Rosnay P, Tavolato C, Thépaut JN, Vitart F (2011) The ERA-interim reanalysis: configuration and performance of the data assimilation system. *Q J R Meteorol Soc* 137(656):553–597. <https://doi.org/10.1002/qj.828>
- DiMarco SF, Reid R (1998) Characterization of the principal tidal current constituents on the Texas-Louisiana shelf. *J Geophys Res* 103(C2):3093–3109
- DiMego G, Bosart L, Endersen G (1976) An examination of the frequency and mean conditions surrounding frontal incursions into the Gulf of Mexico and Caribbean Sea. *Mon Weather Rev* 104(6):709–718
- Eldevik T, Dysthe KB (2002) Spiral Eddies. *J Phys Oceanogr* 32(3):851–869. [https://doi.org/10.1175/1520-0485\(2002\)032<0851:se>2.0.co;2](https://doi.org/10.1175/1520-0485(2002)032<0851:se>2.0.co;2)
- Fox-Kemper B, Ferrari R, Hallberg R (2008) Parameterization of mixed layer eddies. Part I: theory and diagnosis. *J Phys Oceanogr* 38(6):1145–1165. <https://doi.org/10.1175/2007JPO3792.1>
- Hetland RD (2017) Suppression of baroclinic instabilities in Buoyancy-Driven flow over sloping bathymetry. *J Phys Oceanogr* 47(1):49–68. <https://doi.org/10.1175/JPO-D-15-0240.1>
- Hetland RD, DiMarco SF (2012) Skill assessment of a hydrodynamic model of circulation over the Texas–Louisiana continental shelf. *Ocean Model* 43–44:64–76. <https://doi.org/10.1016/j.ocemod.2011.11.009>
- Jaeger GS, Mahadevan A (2018) Submesoscale-selective compensation of fronts in a salinity-stratified ocean. *Sci Adv* 4(2):1–9. <https://doi.org/10.1126/sciadv.1701504>
- Mahadevan A, Tandon A, Ferrari R (2010) Rapid changes in mixed layer stratification driven by submesoscale instabilities and winds. *J Geophys Res Ocean* 115(3):1–12. <https://doi.org/10.1029/2008JC005203>
- Marta-Almeida M, Ruiz-Villarreal M, Pereira J, Otero P, Cirano M, Zhang X, Hetland RD (2013) Efficient tools for marine operational forecast and oil spill tracking. *Mar Pollut Bull* 71(1–2):139–51. <https://doi.org/10.1016/j.marpolbul.2013.03.022>
- McWilliams JC (2016) Submesoscale currents in the ocean. *Proc R Soc A Math Phys Eng Sci* 472(2189):20160117. <https://doi.org/10.1098/rspa.2016.0117>
- Mensa JA, Garraffo Z, Griffa A, Özgökmen TM, Haza A, Veneziani M (2013) Seasonality of the submesoscale dynamics in the Gulf Stream region. *Ocean Dyn* 63(8):923–941. <https://doi.org/10.1007/s10236-013-0633-1>
- Munk W, Armi L, Fischer K, Zachariasen F (2000) Spirals on the sea. *Proc R Soc A Math Phys Eng Sci* 456(1997):1217–1280. <https://doi.org/10.1098/rspa.2000.0560>
- Munoz SE, Dee SG (2017) El niño increases the risk of lower Mississippi River flooding. *Scientific Reports* 7(1):1–7. <https://doi.org/10.1038/s41598-017-01919-6>
- Munoz SE, Giosan L, Therrell MD, Remo JW, Shen Z, Sullivan RM, Wiman C, O'Donnell M, Donnelly JP (2018) Climatic control of Mississippi River flood hazard amplified by river engineering. *Nature* 556(7699):95–98. <https://doi.org/10.1038/nature26145>
- Özgökmen TM, Poje AC, Fischer PF, Haza AC (2011) Large eddy simulations of mixed layer instabilities and sampling strategies. *Ocean Model* 39(3–4):311–331. <https://doi.org/10.1016/j.ocemod.2011.05.006>
- Shchepetkin A, McWilliams J (2005) The regional oceanic modeling system (ROMS): a split-explicit, free-surface, topography-following-coordinate oceanic model. *Ocean Model* 9(4):347–404. <https://doi.org/10.1016/j.ocemod.2004.08.002>
- Simpson J, Sharples J (2012) Introduction to the physical and biological oceanography of shelf seas. Cambridge University Press
- Smolarkiewicz P (2006) Multidimensional positive definite advection transport algorithm : an overview. *Int J Numer methods fluids* (September 2005):1123–1144
- Stefan HG, Preud'homme EB (1993) Stream temperature estimation from air temperature. *JAWRA Journal of the American Water Resources Association* 29(1):27–45. <https://doi.org/10.1111/j.1752-1688.1993.tb01502.x>
- Taylor KE (2001) Summarizing multiple aspects of model performance in a single diagram. *J Geophys Res* 106(D7):7183. <https://doi.org/10.1029/2000JD900719>
- Thomas LN, Tandon A, Mahadevan A (2013) Submesoscale Processes and Dynamics. In: *Ocean Model. an Eddying Regime*, pp 17–38, <https://doi.org/10.1029/177GM04>
- Wang L, Justić D (2009) A modeling study of the physical processes affecting the development of seasonal hypoxia over the inner Louisiana-Texas shelf: Circulation and stratification. *Cont Shelf Res* 29(11–12):1464–1476. <https://doi.org/10.1016/j.csr.2009.03.014>
- Warner J, Sherwood C, Arango H (2005) Performance of four turbulence closure models implemented using a generic length scale method. *Ocean Model* 8(1–2):81–113. <https://doi.org/10.1016/j.ocemod.2003.12.003>
- Willmott C (1982) Some comments on the evaluation of model performance. *Bull Am Meteorol Soc* 63(11):1309–1313
- Zhang X, Hetland RD, Marta-Almeida M, DiMarco SF (2012) A numerical investigation of the Mississippi and Atchafalaya freshwater transport, filling and flushing times on the Texas-Louisiana Shelf. *J Geophys Res* 117(C11):C11009. <https://doi.org/10.1029/2012JC008108>

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